

MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT) — THIRD SEMESTER, NOVEMBER 2023

Paper Code: ES-201

Subject: Computational Methods

Time: 1½ Hrs.

Max. Marks: 30

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Marks	CO
1(a)	Write the statement of Lagrange's Mean Value Theorem and apply it to the function $f(x) = \sqrt{x}$ in $[0, 2]$.	[2.5]	CO1
1(b)	Find minimum value of $f(x) = x^3 - 3x - 7$, using Newton's Method taking initial approximation $x_1 = 0.5$ and $\varepsilon = 0.001$.	[2.5]	CO1
1(c)	Prove that: $\Delta^2 y_3 = \nabla^2 y_5$.	[2.5]	CO2
1(d)	If $f(x) = 1/x$, find divided differences $[a, b]$ and $[a, b, c]$.	[2.5]	CO2
2(a)	Find minimum value of $f(x_1, x_2) = x_1^2 - x_1 x_2 + x_2^2$, using Steepest descent method such that $ f(X_{k+1}) - f(X_k) < 0.05$ taking starting point as $X_1 = (1, \frac{1}{2})$.	[5]	CO1
2(b)	Use Romberg's method to compute $\int_0^1 [dx / (1 + x^2)]$ correct to four decimal places.	[5]	CO2
3(a)	Find the minimum value of $f(x) = x^2 + 2x$ using $n = 4$ within the interval $[-3, 4]$ using Fibonacci Search method.	[5]	CO1
3(b)	Evaluate $\int_0^1 [dx / (1 + x^2)]$ using Simpson's $\blacksquare$ rule taking $h = \frac{1}{4}$ and Simpson's $\blacksquare$ rule taking $h = \blacksquare$.	[5]	CO2
4(a)	Use Secant method to find a root correct to two decimal places, of the equation $x e^x - 2 = 0$ taking two initial approximations as $x_0 = 0.9$ and $x_1 = 1$.	[5]	CO1
4(b)	Find the polynomial of the lowest degree which assumes the values 1, 27 and 64 when x takes the values 1, 3 and 4 respectively, using Lagrange's interpolation formula and hence find $f(2)$.	[5]	CO2

**** End of Question Paper ****